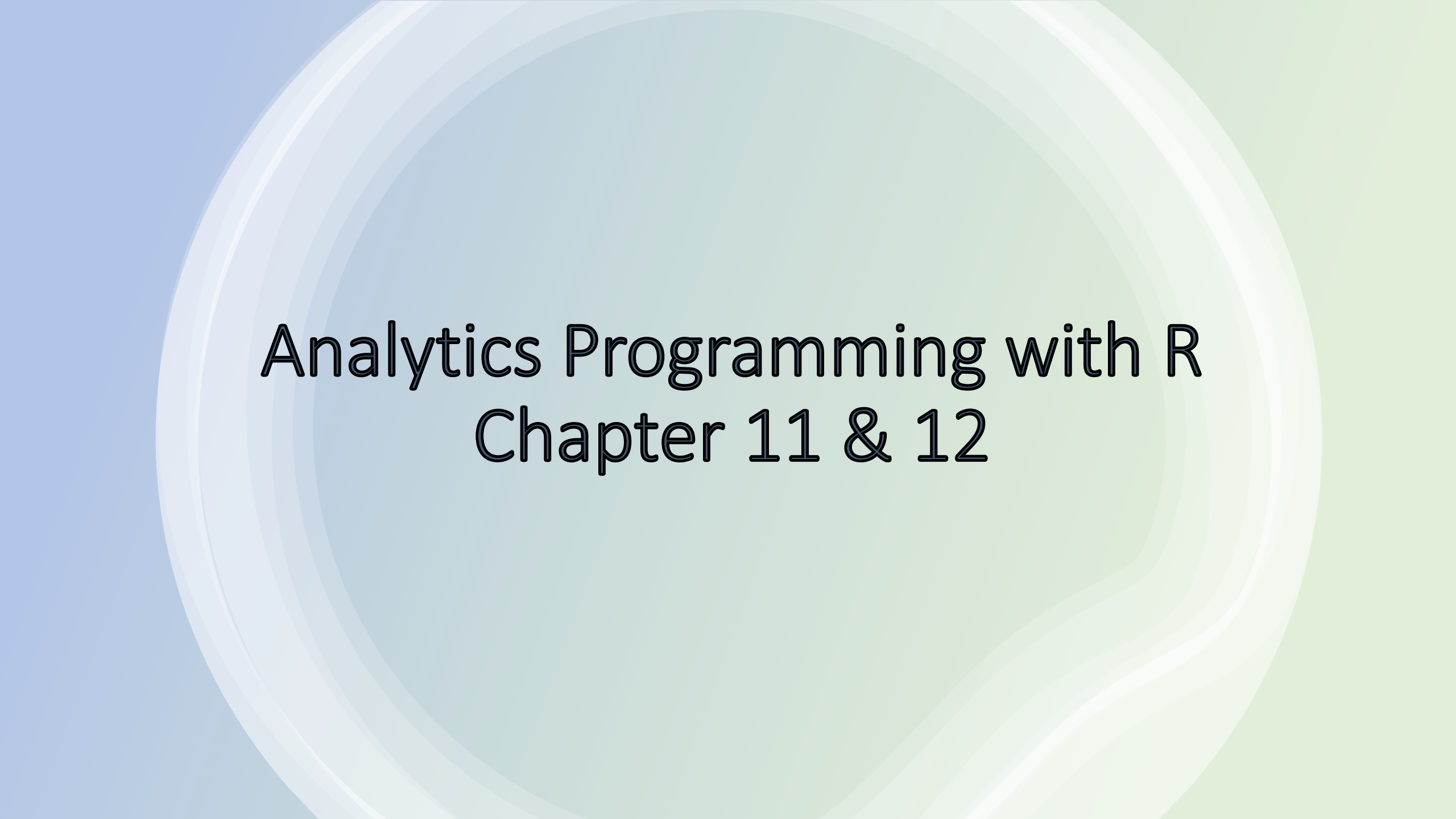




Analytics Programming with R

Chapter 11 & 12

A function for displaying the structure of an object

Function	Description
<code>str(object)</code>	Displays the structure of an object. For a tibble, this includes the column names and data types.

A parameter of the `str()` function

Parameter	Description
<code>strict.width</code>	Specify if the width argument should be followed strictly. Value can be no, wrap, or cut (one line per column).

How to use piping with the `str()` function

```
Data_seatbelt %>% str(strict.width = "cut")
```

Some functions for examining data

```
length(x)
```

```
unique(x)
```

How to get the number of rows in a column

```
length(data_seatbelt)
```

How to get the unique values of a column

```
unique(data_seatbelt$Region)
```

How to get a count of the unique values of a column

```
length(unique(data_seatbelt$Region))
```

How to use piping instead of nested functions

```
data_seatbelt$Region %>% unique() %>% length()
```

A function for applying another function to all columns

`apply(X, MARGIN, FUN)`

A function for sorting data

Function	Description
<code>arrange(.data, ...)</code>	Sorts the rows of a data frame by the specified column or columns. To sort in descending order, enclose the column in the <code>desc()</code> function.

A parameter of the `arrange()` function

Parameter	Description
<code>.by_group</code>	Specifies whether to sort by groups. Default is <code>FALSE</code> .

How to sort the data by a column from smallest to largest

```
fires %>% arrange(FIRE_SIZE)
# A tibble: 49,797 x 5
  FIRE_NAME      FIRE_SIZE STATE FIRE_YEAR DISCOVERY_DATE
  <chr>          <dbl> <chr>    <int> <chr>
1 NA              100. GA         2007 2007-02-13 00:00:00
2 NADINE NAFF      100. WA         2006 2006-06-09 00:00:00
3 NA              100. SC         2002 2002-03-01 00:00:00
...
```

How to sort the data from largest to smallest

```
fires %>% arrange(desc(FIRE_SIZE))
# A tibble: 49,797 x 5
  FIRE_NAME      FIRE_SIZE STATE FIRE_YEAR DISCOVERY_DATE
  <chr>          <dbl> <chr>    <int> <chr>
1 INOWAK        606945 AK         1997 1997-06-25 00:00:00
2 LONG DRAW     558198. OR         2012 2012-07-08 00:00:00
3 WALLOW        538049 AZ         2011 2011-05-29 00:00:00
...
```

How to sort by multiple columns

```
fires %>% arrange(STATE, FIRE_YEAR, desc(FIRE_SIZE))
# A tibble: 49,797 x 5
  FIRE_NAME  FIRE_SIZE STATE FIRE_YEAR DISCOVERY_DATE
  <chr>      <dbl> <chr>    <int> <chr>
1 213225      48087 AK        1992 1992-06-11 00:00:00
2 232414      35090 AK        1992 1992-07-19 00:00:00
3 BTT 3 48      25600 AK        1992 1992-06-23 00:00:00
...
```

How to sort by multiple columns with groups

```
fires %>% group_by(STATE) %>%
  arrange(FIRE_YEAR, desc(FIRE_SIZE), .by_group = TRUE)
# A tibble: 49,797 x 5
# Groups:   STATE [50]
  FIRE_NAME  FIRE_SIZE STATE FIRE_YEAR DISCOVERY_DATE
  <chr>      <dbl> <chr>    <int> <chr>
1 213225      48087 AK        1992 1992-06-11 00:00:00
2 232414      35090 AK        1992 1992-07-19 00:00:00
3 BTT 3 48      25600 AK        1992 1992-06-23 00:00:00
...
```

Two functions for selecting rows

```
filter(.data, ...)  
unique(data)
```

Some logical operators

Operator Name	
&	AND
	OR
!	NOT

How to filter for one condition

```
polls %>% filter(state != "U.S.")
```

How to filter for multiple conditions

```
polls %>% filter(state == "California" & grade == "A")
```


Some logical functions

Function	Name
<code>%in%</code>	IN
<code>xor(x, y)</code>	XOR
<code>duplicated(x)</code>	NOT

How to filter for a value in a vector of values

```
polls %>% filter(state %in% c("California", "Oregon", "Washington"))
```

How to view duplicate rows

```
fires %>% filter(duplicated(fires))
# A tibble: 441 x 6
  FIRE_NAME FIRE_SIZE STATE FIRE_YEAR DISCOVERY_DATE    CONTAIN_DATE
  <chr>      <dbl> <chr>    <int> <chr>          <chr>
1 NA         1500 CO      2005 2005-07-24 00:00:00 NA
2 NA         250 FL      2004 2004-06-14 00:00:00 NA
3 NA         125 KY      2003 2003-03-28 00:00:00 NA
...
```

How to drop duplicate rows

```
fires <- fires %>% filter(!duplicated(fires))
```

Another way to drop duplicate rows

```
fires <- fires %>% unique()
```

A function for selecting columns

```
select(.data, ...)
```

The unique polls data for the first six columns

..\$ cycle	: int 1	One unique value: Delete?
..\$ branch	: int 1	
..\$ type	: int 3	Small number of unique values: Often groups or categories.
..\$ matchup	: int 1	
..\$ forecastdate	: int 1	
..\$ state	: int 57	

How to use indexes to select columns

```
# col with specified index  
polls %>% select(1)
```

```
# all cols in specified range  
polls %>% select(1:5)
```

```
# each col with a specified index  
polls %>% select(1, 2, 4)
```

```
# all cols except specified indexes  
polls %>% select(-1, -2, -4)
```

```
# another way to code previous  
polls %>% select(-c(1, 2, 4))
```

How to use names to select columns

```
# specified column name  
polls %>% select(cycle)
```

```
# all cols in specified range  
polls %>% select(cycle:forecastdate)
```

```
# each col with a specified name  
polls %>% select(cycle, branch, matchup)
```

```
# all cols except specified names  
polls %>% select(-cycle, -branch, -matchup)
```

```
# more ways to code previous  
polls %>% select(-c(cycle, branch, matchup))  
polls %>% select(-c("cycle", "branch", "matchup"))
```

Functions for working with missing values

`complete.cases(...)`

`is.na(col)`

`nrow(data)`

`replace_na(col, val)`

`ifelse(test, yes, no)`

Functions for changing a column's data type

```
mutate(.data, ...)
```

```
as.integer(x)
```

```
as.double(x)
```

Functions for converting strings to dates and times

`as.Date(x, format)`

`as_date(x, format)`

`as_datetime(x, format)`

Some date/time format specifiers

Code	Description
%d	Numeric day (3, 17, 31).
%a	Abbreviated name of day (Mon, Wed, Fri).
%m	Numeric month (1, 6, 12).
%b	Abbreviated name of month (Jan, Feb, Mar).
%y	Abbreviated year (96, 02, 22).
%Y	Full year (1996, 2002, 2022).
%H	Hours
%M	Minutes
%S	Seconds